
An investigation into question-based methods for uncertainty detection in LLMs

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Abstract

Recent works on detecting hallucinations by large language model (LLM) responses have used token probability or logits produced by repeated prompting to probe for uncertainty. In this work, we propose an expansion of such approaches using LLM-generated question paraphrases. We experiment with using this approach to extend to existing uncertainty metrics: semantic entropy and semantic energy. Our results show a small improvement over the original methods. We also run an ablation study that suggests that checking paraphrase equivalence to the original question is a useful step.

1 Introduction

1.1 Problem definition

The problem we will be investigating is estimating LLM response confidence and detecting hallucination. Hallucinations are instances of a large language model expressing plausible but false information. Prior research suggests that these errors are the natural outcomes of training that rewards guessing over a lack of response or acknowledgment of uncertainty [5]. Being able to successfully automatically label an LLM’s uncertainty in responding to a prompt could provide a useful metric to inquire about an LLM’s pre-training knowledge, implement truthfulness alignment training, or directly help users evaluate the usefulness/truthfulness of a response. A reliable confidence score could increase trust in high-scoring responses. Adversarial or out-of-distribution prompts that make a model fabricate false information can be very harmful in high stakes contexts, such as in medical and legal AI. A measure to detect hallucination in the response of a model and warn users or prevent the model from responding with hallucinated information would help reduce risk.

1.2 Relevance to trustworthy aspects

Detecting hallucination is relevant to robustness, explainability, and even adversarial attacks. A common notion of robustness in machine learning is how stable model performance is in response to variations in its input. This has been measured by the accuracy and variance of model responses in ϵ -perturbed training inputs. Accordingly, if a model generates varying and inaccurate responses for the same prompt, or multiple semantically similar prompts, as in hallucination, we may claim it is not robust. Detecting hallucination is also key to explainability. Clearly, fabricating or contradicting factual information, such as “Sarah Collins was the first person to travel to Mars in 2022”, or being logically or contextually inconsistent, such as Q: “Translate ‘plastic pollution harms marine life’ to Arabic” and A: “Plastic is commonly found in the ocean”, detracts from a model’s explainability (CITE). Furthermore, a reliable confidence score in itself, like an explanation, would serve as a useful piece of meta-information about a model response and could increase trust in high-scoring responses. Finally, adversarial attacks that attempt to make a model fabricate false information can

be very harmful in high stakes contexts, such as in medical and legal AI. A defense where we can detect hallucination in a model’s response and stop the model from responding with hallucinated information would help deter such attacks. Through hallucination detection, we plan to evaluate these trustworthy aspects, like quantifying how robust a Question-Answer (QA) LLM is. We may possibly apply or extend them based on the results our initial investigations.

2 Related works

To help guide hallucination research, Alansari and Luqman [1] give an overview of the paradigms in the literature and offer their own framework for kinds and causes of hallucination and hallucination detection. They claim that individual hallucination detection techniques do not work well under all circumstances. For example, retrieval-based methods are highly sensitive to the quality of external data, and uncertainty-based methods often fail to detect hallucinations when the model shows high confidence in an incorrect response. This work guides our suggested approach for investigation, and inspires us to try out and investigate composite techniques.

The basis for our project is Farquhar et al. [1]’s semantic entropy. This uncertainty-based measure of hallucination uses the principle that high variability in model responses indicates high likelihood of confabulation in those responses. Thus, they construct the measure of semantic entropy – entropy calculated over multiple sampled responses clustered by semantic equivalence, achieving state-of-the-art AUROC, rejection accuracy, and AURAC metrics compared with other methods and baselines.

Semantic energy [5] is another measure of hallucination that builds on the core idea of semantic entropy. Instead of using post-softmax probabilities to compute entropy, Ma et al. directly use logits, claiming they are a better representation of a model’s uncertainty than the model responses used in semantic entropy, to compute a Boltzmann-based energy measure of semantic clusters. They are able to achieve improvements of 13% in AUROC compared to semantic entropy in cases where semantic entropy is confident. We aim to investigate and expand on semantic entropy and semantic energy.

Another method, ChainPoll [2], repeatedly queries another LLM (GPT-3.5-turbo) about whether the assessed response contained hallucinations with a “detailed CoT” prompt, taking the average of the responses (0 or 1). ChainPoll does not require any information beyond the GPT-3.5-turbo responses, so it can run on black-box models. While ChainPoll achieved a similar performance to semantic entropy on detecting hallucinations in TriviaQA, it seems to rely on GPT-3.5-turbo’s ability to ascertain the correctness or feasibility of the given answer. This method would likely not be robust to hallucinations/questions that are out of distribution for GPT-3.5-turbo. While semantic entropy does use an entailment model to predict semantic equivalence between responses, entailment is a simpler task and the determination of hallucination is more dependent on the entropy among the primary model’s responses. We are also unable to test ChainPoll since their chain-of-thought prompt does not appear to be publicly available.

Xiong et al. [7] used a similar approach, but instead of using a separate LLM and entirely new prompt to detect hallucinations, they alter the desired prompt to elicit an answer along with a confidence level and test several different aggregation methods for the resulting answers and confidences. However, they found that models were over-confident and that a more effective predictor was the consistency of a model’s responses when sampled multiple times. This is similar to semantic entropy but looked for perfect matches rather than semantic equivalence among responses and would be particularly unfit for assessing confidence in longer answers.

3 Methods

We investigate semantic entropy (SE) [1] and semantic energy (SEnergy), methods for quantifying LLM hallucination and uncertainty. Given the token probabilities, SE can be applied to any LLM and is calculated over information retrieval datasets, such as TriviaQA and SQuAD. In the black-box case, we cannot directly compute SE but can compute “discrete” semantic entropy, further explained below.

We can apply SEnergy in the same situations, except that we need access to the model logits.

The LLMs we evaluate is Llama-2-7b-chat downloaded from HuggingFace, the entailment model we use is DeBERTa-V2-XLarge-MNLI, and the dataset we use is TriviaQA.

We compute SE and SEnergy over 1) multiple answers to a single prompt and 2) multiple answers to multiple prompts. In case 1, we generate 10 high temperature answers and 1 low temperature answer, and in case 2, we generate 3 additional paraphrasings, which may be cut in the case of non-equivalence, of the original question, and generate 3 high temperature answers and 1 low temperature answer for each of these questions. The SE is computed over the high temperature answers.

We use AUROC as the primary metric to evaluate our hallucination prediction methods. AUROC is the area under the receiver-operating characteristic curve (ROC), which plots true positive rate (sensitivity) over false positive rate. This is a common metric for binary classification and is typically a better metric than accuracy for unbalanced datasets, since it summarizes the performance of a model over all possible thresholds for a continuous probability estimation, in this case semantic entropy.

3.1 Semantic entropy

SE as given in [1] is calculated as follows:

1. Generate sequences $\mathbf{s}$ given a context $\mathbf{x}$.
2. Cluster the generated sequences by semantic equivalence.
3. Estimate the entropy over the equivalence classes found in step 2.

Recall that the entropy of a random variable Y conditioned on an observation X is $H(Y|X) = -\sum_y P(y|X) \log P(y|X)$, where y is a possible realization of Y .

Since token probabilities are conditionally independent, the conditional probability of a sequence $\mathbf{s}$ given its context $\mathbf{x}$ is the product of the conditional probabilities of the tokens in $\mathbf{s}$ given previous tokens,

$$P(\mathbf{s}|\mathbf{x}) = \prod_i P(s_i|s_{<i}, \mathbf{x}),$$

where s_i is the i -th token of $\mathbf{s}$. Thus,

$$\log P(\mathbf{s}|\mathbf{x}) = \sum_i \log P(s_i|s_{<i}, \mathbf{x}).$$

Furthermore, we length-normalize $\log P(\mathbf{s}|\mathbf{x})$ since longer sentences have exponentially smaller joint probabilities in expectation, and thus linearly smaller log probabilities,

$$\frac{1}{N} \sum_i \log P(s_i|s_{<i}, \mathbf{x}).$$

Let C denote the set of equivalence classes we compute from step 2. Let $c \in C$ denote a single equivalence class. Then, we define

$$P(c|\mathbf{x}) = \sum_{\mathbf{s} \in c} P(\mathbf{s}|\mathbf{x}).$$

Thus, the entropy over the equivalence classes is estimated by

$$H(C|\mathbf{x}) = -\sum_{c \in C} P(c|\mathbf{x}) \log P(c|\mathbf{x}).$$

This is an estimation since C is not necessarily the set of all possible equivalence classes we could have given $\mathbf{x}$.

In the black-box setting, we can compute discrete SE by letting $P(c|\mathbf{x}) = \sum_{\mathbf{s}} \frac{I_{\{\mathbf{s} \in c\}}}{|C|}$, e.g. the proportion of all sampled answers that belong to cluster c .

3.2 Semantic energy

Semantic energy (SEnergy) aims to capture uncertainty in cases where the probability distribution and semantic result are the same as when there is little uncertainty, but where the logits before softmax differ. [szhou: INSERT IMAGE]

Semantic energy (SEnergy) as given in [5] is calculated as follows:

1. Steps 1+2 of semantic entropy (generating & clustering sequences)
2. Calculate the SEnergy for each of the equivalence classes found in step 2. These values will later be used in the calculation of AUROC. The SEnergy over all clusters is simply taken as the maximum SEnergy that was computed for each cluster.

For a given sequence $\mathbf{s}$, let s_i indicate the i -th token of $\mathbf{s}$. Let $T_{\mathbf{s}}$ be the total number of tokens in $\mathbf{s}$. Let $z(s_i)$ indicate the logit value of s_i . The “average energy” of a sequence $\mathbf{s}$ is approximated as $E(\mathbf{s}) \approx \frac{1}{T_{\mathbf{s}}} \sum_{s_i \in \mathbf{s}} -z(s_i)$. Then, the SEnergy of a single cluster is very simply calculated as:

$$U(c) = \sum_{\mathbf{s} \in c} E(\mathbf{s}).$$

The lower $U(c)$, the less uncertainty for that cluster.

3.3 Generating Paraphrased Questions

We generate paraphrases for questions in the tested information retrieval datasets by prompting Llama-2-7b-chat. To generate n paraphrase candidates for a given question, we input the following prompt:

"For the following question Q, please provide n paraphrases of the question that are semantically equivalent to it. Q: *[question]* Paraphrased questions:"

In order to ensure that the provided paraphrased questions are valid substitutes, we check for semantic equivalence between the original question-answer pair and each paraphrased question-answer pair (using the original reference answer for both). We include the answer in each so that we can determine semantic equivalence by bidirectional entailment. For example, "Who was the first president of the United States? George Washington" and "What was the name of the first president of the United States? George Washington" entail each other. We keep all questions that are determined to be effectively equivalent to the original question and use these for our semantic entropy variants by generating several high temperature responses to the original question and each accepted paraphrase. For our entailment model, we used DeBERTa (deberta-v2-xlarge-mnli) [3].

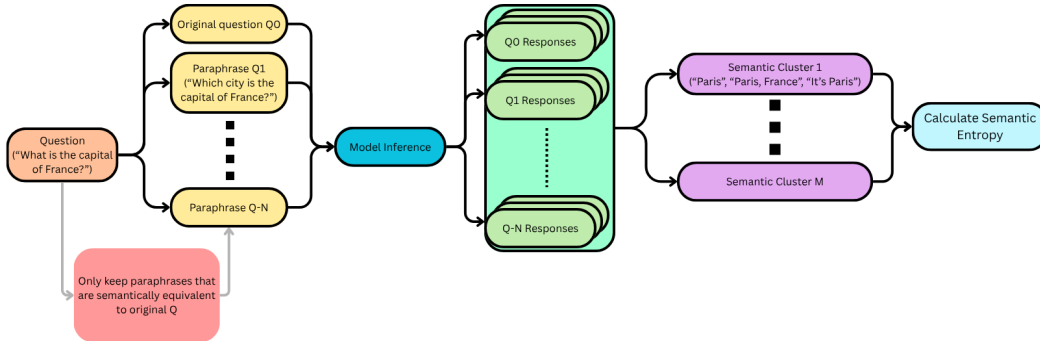


Figure 1: Paraphrase-Augmented Semantic Flowchart.

4 Experiments

4.1 Original and Paraphrase-Augmented SE on TriviaQA

We tested our paraphrase-augmented semantic entropy and compared it with the original method and a baseline: $P(\text{false})$ (with and without paraphrase responses).

$P(\text{false})$ is 1 minus the probability the model assigns to "A" (True) when prompted to evaluate whether its predicted response (with temperature=0.1) to a question is true/false given a list of "brainstormed" answers (model outputs with temperature=1.0). We also used a paraphrase-augmented $P(\text{false})$ that used brainstormed responses aggregated across the original question and all valid paraphrases.

We run our experiments with the TriviaQA dataset [4], which consists of 95k question-answer pairs. TriviaQA provides aliases for answers, giving us a semantic ground truth to compare to LLM responses in order to assess accuracy / hallucination.

Due to time constraints, we calculated metrics for a subset of 1095 original TriviaQA questions. For roughly half (507) of the samples, we were able to generate three valid paraphrases, while for 195, all three generated paraphrases were labeled as invalid (not bidirectionally entailed with original). Some main results of our exploratory data analysis of the sample data as well as our generated paraphrases are shown in Figure 2.

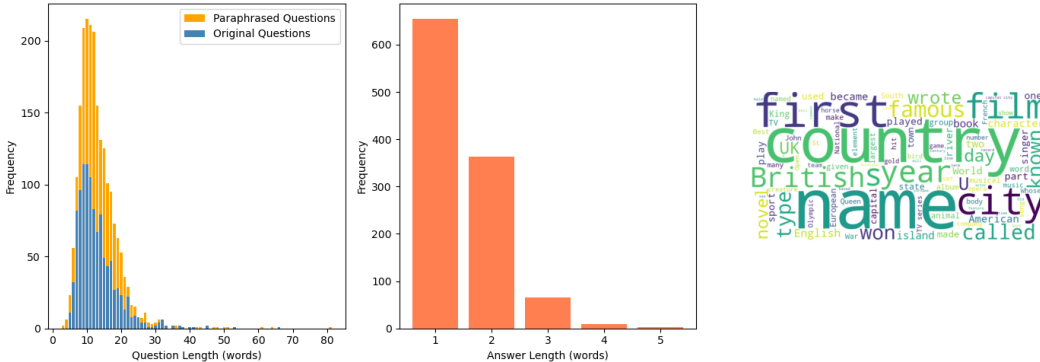


Figure 2: Left: Distributions of original question and paraphrase question lengths (in words). Center: Distribution of reference answer lengths. Right: Word cloud showing most frequent words in original questions.

For the tested subset of TriviaQA, our paraphrase-augmented semantic entropy achieved an AUROC of 0.820 compared to regular semantic entropy’s 0.817. Both versions of semantic entropy (SE) significantly outperformed both versions of $P(\text{false})$ – regular $P(\text{false})$ had AUROC 0.767 and paraphrase-augmented $P(\text{false})$ had AUROC = 0.766. Results are plotted in Figure 3. We also investigated the effect of the number of paraphrases/questions used, since this varied across samples, and saw an upward trend of AUROC for paraphrase-augmented semantic entropy on examples that had at least one paraphrase, with an exception for questions with no paraphrases. However, this is not a direct comparison since we did not test the same questions with different number of paraphrases included. The results by number of paraphrases are shown in the leftmost plot in Figure 4. The figure also includes two plots showing the spread of regular and paraphrase-augmented semantic entropy for correctly versus incorrectly predicted questions. The paraphrase-augmented SE has a higher spread for incorrectly predicted samples, although there also appears to be more rightward skew for correctly predicted samples. Also, noticeably more correct predictions are in the first bucket (< 0.25) for paraphrase-augmented SE compared to regular SE, suggesting some improved discriminative behavior.

4.2 Ablation: Checking paraphrase equivalence

We investigated the importance of our paraphrase equivalence check step, where we only keep generated paraphrases that are bidirectionally entailed with the original question (with answer suffix added to both). To do so, we reran our experiment on a sample of 600 and kept a list of all generated

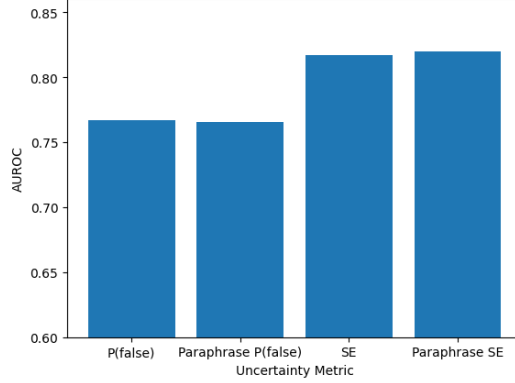


Figure 3: AUROC for Original P(false) and Semantic Entropy compared with Paraphrase-Augmented versions

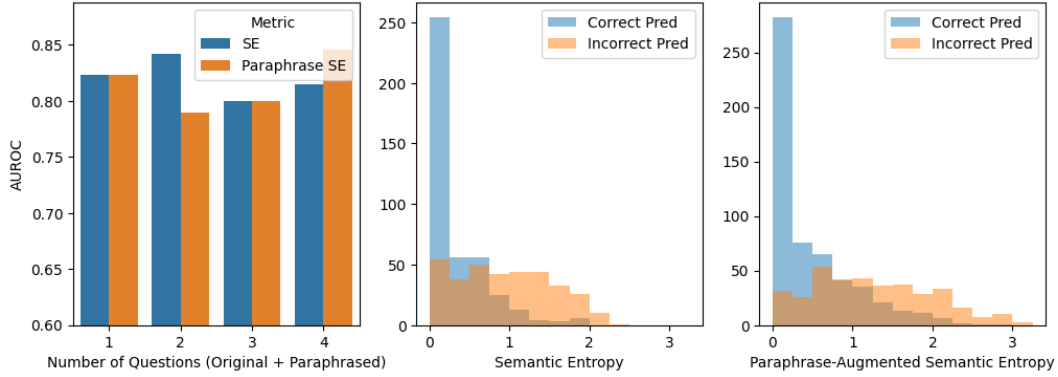


Figure 4: Comparisons of regular and paraphrase-augmented semantic entropy on TriviaQA. Left: AUROCs for both metrics on samples grouped by the number of valid (equivalent) generated paraphrases (1 = only original question in which case SE and P-A SE are equal). Center: Histogram comparing regular SE of correctly and incorrectly predicted samples. Right: Correct/incorrect histogram for paraphrase-augmented SE.

paraphrases along with our original list of validated paraphrases. We ended up with 579 usable samples after filtering out samples for which Llama did not successfully generate three paraphrases. We then ran the following steps on both sets and compared our final results. The final results showed a somewhat higher AUC when entailment was required ($AUC=0.852$) compared to when all generated paraphrases were used ($AUC=0.835$).

4.3 Comparison with SEnergy

We also investigated SEnergy as an alternative uncertainty estimation method that we can compute over paraphrase-augmented datasets. However, since SEnergy is originally given as a sum of negative means of sequences of logits, SEnergy scales by the numbers of sequences and thus is not robust over different cluster sizes. For example, supposing the average negative mean of logits is ≈ 20 , when there are no additional paraphrasings of a question, there will only be one cluster and the SEnergy score for that question will be 20. When there are three additional paraphrasings of a question in a single cluster, the SEnergy score for that question will be 80.

As such, when computing SEnergy over the augmented dataset, we divide the cluster SEnergy scores by the number of total answers for the current question. This “normalizes” the scale of SEnergy scores when there are different numbers of responses.

In figure 5, we ran uncertainty detection on an unmodified subset of TriviaQA, and saw that SE achieves an AUROC score of 0.760 while SEnergy achieves an AUROC score of 0.769. We also

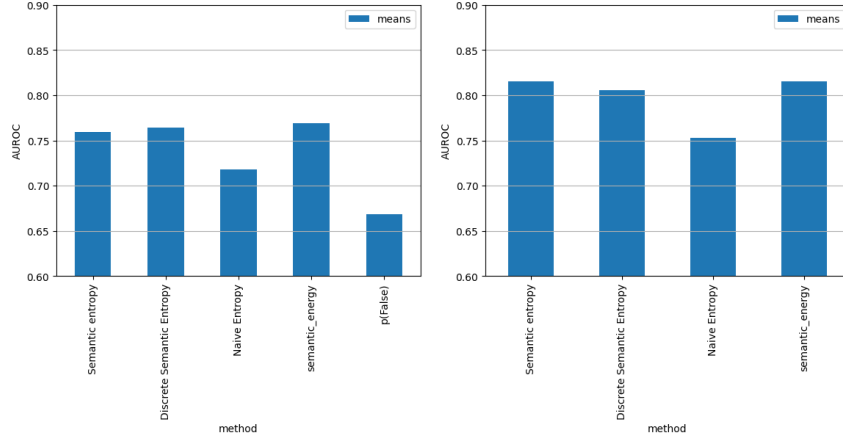


Figure 5: Mean AUROC scores for SEnergy, SE, p(False), and other methods on a non-paraphrase-augmented subset of TriviaQA (left) and a paraphrase-augmented subset of TriviaQA (right) using Llama-2-7b-chat

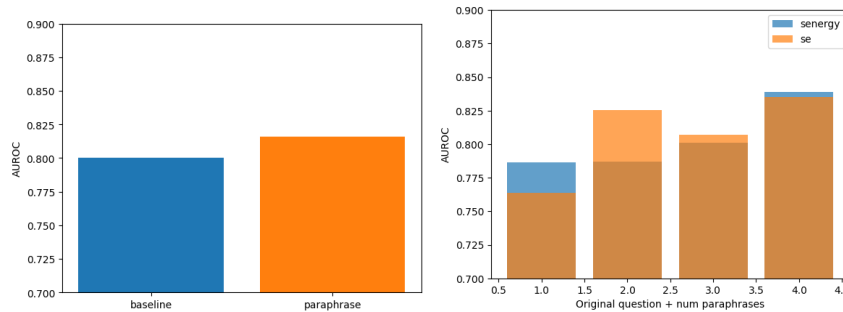


Figure 6: Baseline vs paraphrase augmentation comparison for SEnergy (left), and SE vs SEnergy comparison for different numbers of questions in the paraphrase-augmented case(right)

ran uncertainty detection on a paraphrase-augmented subset of TriviaQA using normalization for SEnergy, where SE achieved an AUROC of 0.816 and SEnergy also achieved an AUROC of 0.816. In general, the performance of SE and SEnergy are very similar, and SEnergy at least as well as SE on the data that we experiment on.

In figure 6, we compare the effect of paraphrasing on SEnergy with the baseline answer generation method. SEnergy with paraphrases obtains an AUROC of 0.816 while baseline SEnergy obtains an AUROC of 0.8. Again, augmenting with paraphrases performs just a bit better than the baseline. As in section 4.2, we also grouped the SE and SEnergy computations in the paraphrase-augmented case for different numbers of questions. SE exhibits an interesting small trend where it achieves a local peak when there are two total questions, while SEnergy monotonically increases in AUROC as the total number of questions increases.

In figure 7, we create histograms of the number of incorrect and correct responses in each bucket of SEnergy values in the case with paraphrasing and paraphrasing and normalization and in the original method. From a qualitative visual inspection, we can see that the densities of SEnergy appear more spread out than that of SE, and that incorrect answers have more density on the right tail of the SEnergy distribution than in SE. This aligns with the results from [5]. It is difficult however to tell if there is a difference in distribution between the case with paraphrasing or without.

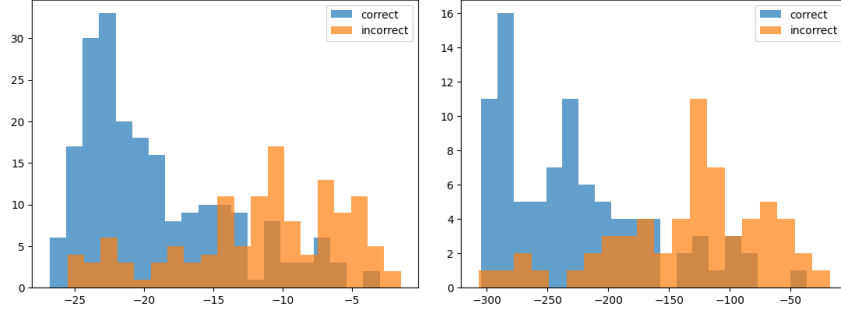


Figure 7: Histograms comparing correct/incorrect responses on SEnergy with paraphrasing and normalization (left), and on vanilla SEnergy (right). The horizontal axis is SEnergy score.

5 Conclusion

In our experiments, we found that computing semantic entropy from responses to the original question augmented with responses to paraphrased questions marginally improved the discriminative power for incorrect vs. correct predictions. After normalizing for the number of responses, we found that incorporating these paraphrases also slightly improved results for semantic energy.

One direction for future work would be to test this method on more data and different dataset. Due to time constraints, we used a set of about 1100 examples from TriviaQA. For future experiments, we could expand this sample size as well as test performance on other QA datasets, including more direct retrieval based datasets that may allow us to more directly assess hallucination. For example, the Stanford Question Answering Dataset 2.0 (SQuAD 2.0) includes over 50,000 unanswerable questions [6] that we would expect models to exhibit high uncertainty scores for.

6 Code availability

Our code repository, forked from a semantic uncertainty repository that implements regular semantic entropy, is available at https://github.com/szhou0202/semantic_uncertainty. The augmented semantic entropy pipeline is implemented in the main ("master") branch, and semantic energy is implemented in the "szhou" branch.

7 Individual contributions

7.1 Sivan

I worked on generating the paraphrases and checking for bidirectional entailment with the original question. I worked on getting the regular vs. paraphrase-augmented semantic entropy for the final results. I also reran the experiment with + without entailment checking for an ablation test.

7.2 Sophia

I worked on creating the paraphrase-augmented workflow, adjusting for a different dataset schema than the original repo. I integrated the code for calculating SEnergy and generated the SEnergy results.

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